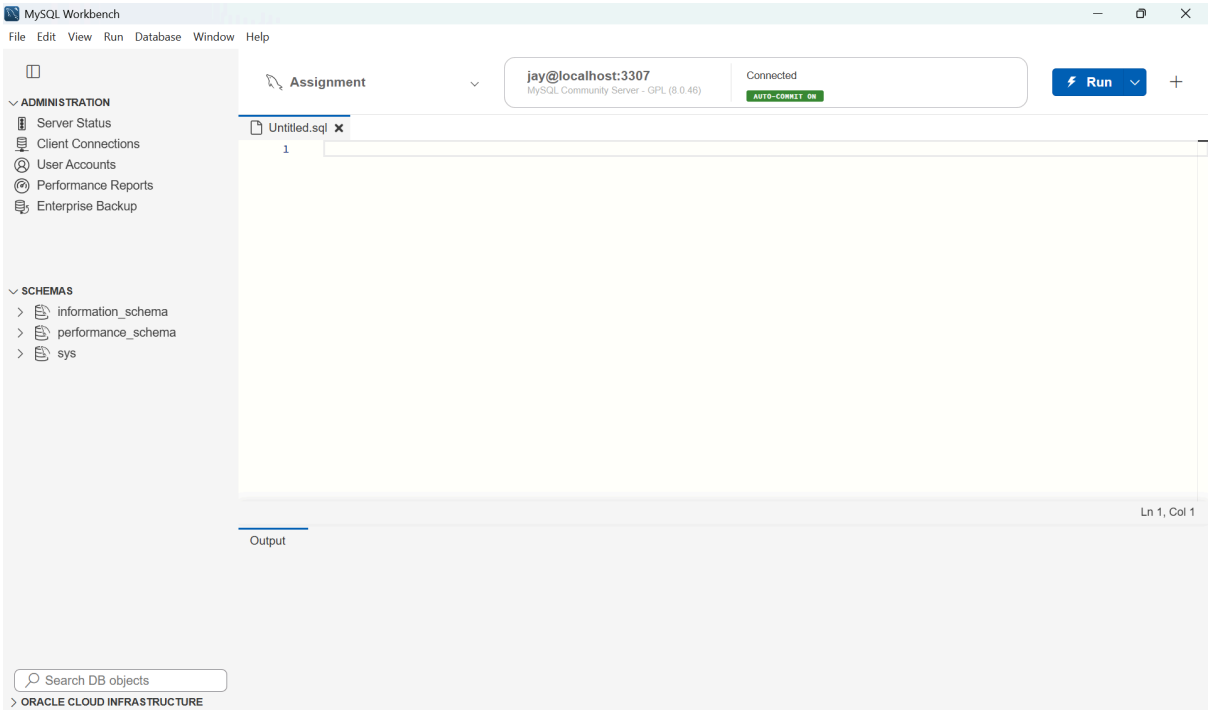
Session - 1

1. Install MySQL Workbench on your computer, connect to the local MySQL server, and take a screenshot of the default databases shown in the Schemas panel.

➔

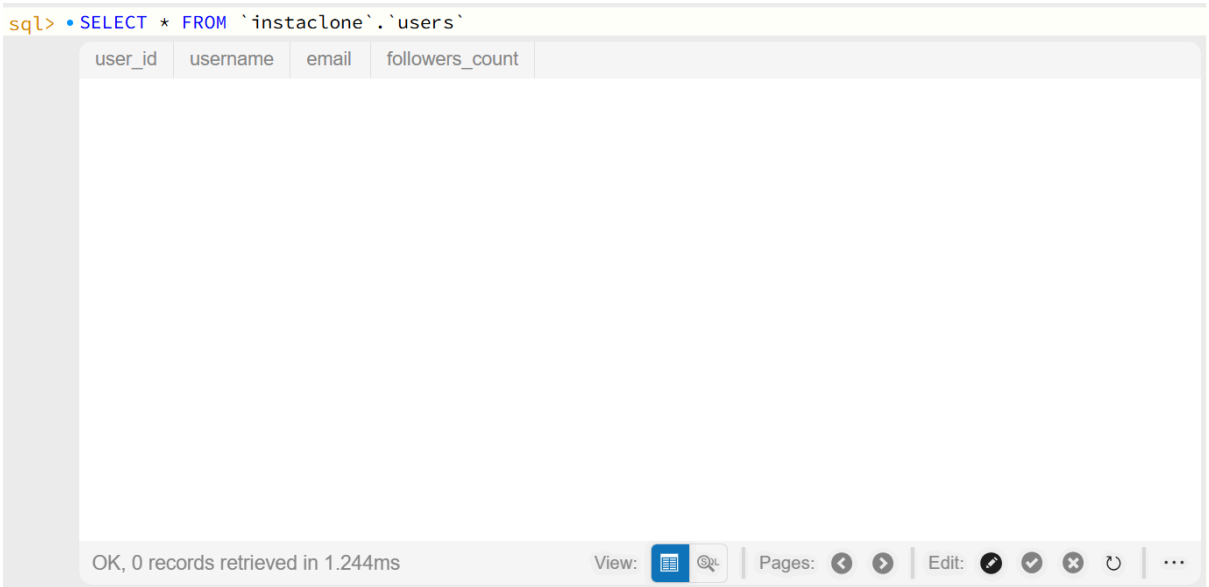


2. Create a new database called InstaClone in MySQL Workbench, then create a table named Users with columns: user_id (INT, primary key), username (VARCHAR), email (VARCHAR), and followers_count (INT).

➔

```
CREATE DATABASE InstaClone;


use instaclone;
CREATE TABLE User(
        user_id INT(11) PRIMARY KEY AUTO_INCREMENT,
        username VARCHAR(255),
        email VARCHAR(255),
        followers_count INT(11)
);
```

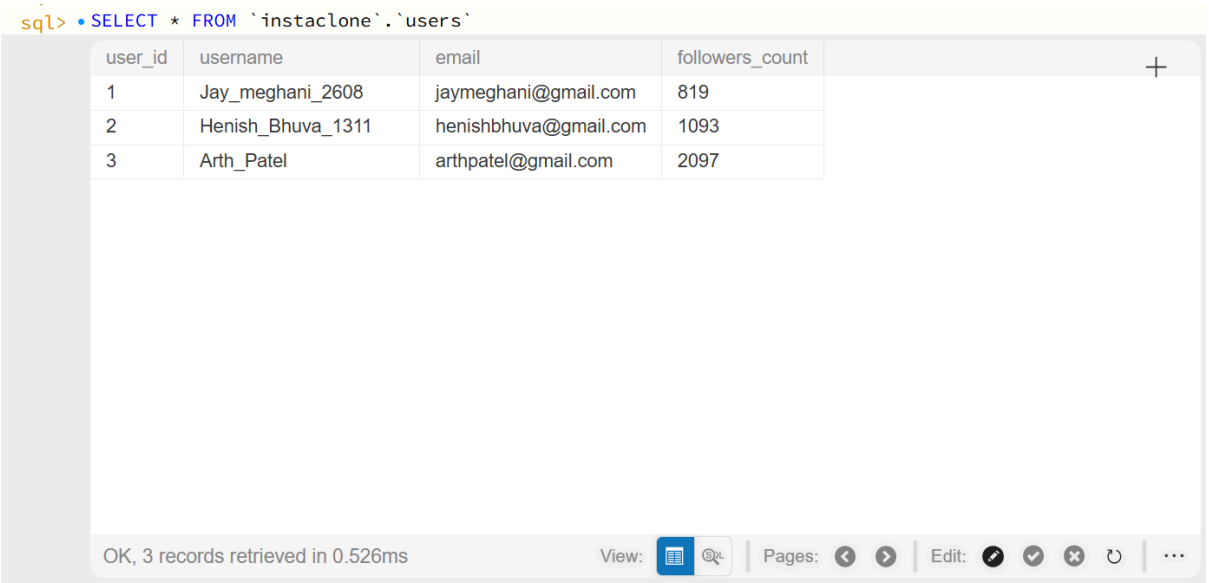


3. Insert 3 sample users into the Users table you created, using realistic Instagram-style usernames and follower counts.

➔

```
use instaclone;

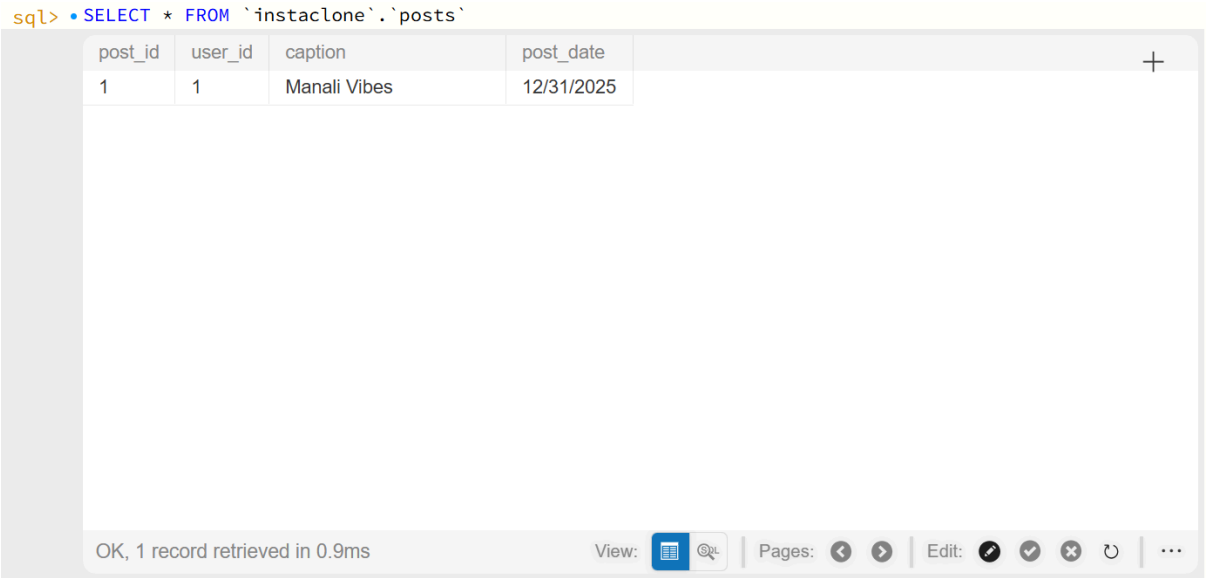
INSERT INTO users(username,email,followers_count)
VALUES
('Jay_meghani_2608','jaymeghani@gmail.com',819),
('Henish_Bhuva_1311','henishbhuva@gmail.com',1093),
('Arth_Patel','arthpatel@gmail.com',2097);
```



4. Create another table in the InstaClone database called Posts with columns: post_id (INT, primary key), user_id (INT), caption (VARCHAR), and post_date (DATE). Add a foreign key from Posts.user_id to Users.user_id to establish a relationship.

➔

```
USE instaclone;
CREATE TABLE Posts(
    post_id INT(11) AUTO_INCREMENT PRIMARY KEY,
    user_id INT(11),
    caption VARCHAR(255),
    post_date DATE,
    FOREIGN KEY (user_id) REFERENCES Users(user_id)
);
INSERT INTO posts(user_id,caption,post_date)
VALUE(1,'Manali Vibes','2025-12-31')
```



5. Write a short comparison (3-4 lines each) of MySQL, PostgreSQL, Oracle, and SQLite, focusing on where each one is commonly used (for example: web apps, mobile apps, enterprise systems, etc.).

➔

MySQL: Open-source, easy to use. Common in web apps and websites (WordPress, e-commerce).

PostgreSQL: Open-source, feature-rich and reliable. Common in data-heavy apps, analytics and fintech.

Oracle: Paid and powerful. Common in large enterprise systems like banks, telecom and government.

SQLite: Lightweight, single-file database with no server. Common in mobile apps and small desktop apps.